

Git Challenges

Resolve Merge Conflicts

- a. Create a merge conflict intentionally (two users editing the same line).

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git merge release
Auto-merging techie
CONFLICT (add/add): Merge conflict in techie
Automatic merge failed; fix conflicts and then commit the result.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Changes to be committed:
  modified:   alphabet
  new file:   release

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both added:   techie

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ !
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet  hello  hotfix  master

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat alphabet
hello world

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git checkout release
Switched to branch 'release'

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (release)
$ ls
hello  release
```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ vi techie

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Changes to be committed:
  modified:   alphabet
  new file:   release

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both added:   techie

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ git add .

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|MERGING)
$ git commit -m "added"
[master 7c2a160] added

```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
alphabet  hello  hotfix  master  release  techie

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat alphabet
hello world

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat techie
hello world and techie

```

1. Resolve the conflict and push the changes.

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat techie
hello world and techie

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "resloved merge conflict"
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
  (commit or discard the untracked or modified content in submodules)
    modified:   techie-05 (untracked content)

no changes added to commit (use "git add" and/or "git commit -a")

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Everything up-to-date
branch 'master' set up to track 'origin/master'.

```

2. Recover Deleted Branch

```
c433502 HEAD@{34}: checkout: moving from hotfix to release
d165cf1 (hotfix) HEAD@{35}: checkout: moving from release to hotfix
c433502 HEAD@{36}: checkout: moving from master to release
0a1c500 HEAD@{37}: commit: alphabet
5f133cd HEAD@{38}: commit: hello
83d91b0 HEAD@{39}: reset: moving to HEAD
83d91b0 HEAD@{40}: commit (amend): changed the commit message using ammend command
0b0dcfc HEAD@{41}: cherry-pick: added in hotfix
ed1ed72 HEAD@{42}: checkout: moving from hotfix to master
d165cf1 (hotfix) HEAD@{43}: checkout: moving from master to hotfix
ed1ed72 HEAD@{44}: checkout: moving from hotfix to master
d165cf1 (hotfix) HEAD@{45}: checkout: moving from master to hotfix
ed1ed72 HEAD@{46}: checkout: moving from hotfix to master
d165cf1 (hotfix) HEAD@{47}: commit: added in hotfix
c433502 HEAD@{48}: merge release: Fast-forward
e49c7d2 HEAD@{49}: checkout: moving from release to hotfix
c433502 HEAD@{50}: commit: added n=in release
```

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git merge hotfix
Merge made by the 'ort' strategy.

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git branch -d hotfix
Deleted branch hotfix (was d165cf1).

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

3. Delete a local branch and then recover it using the reflog.

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git merge hotfix
Merge made by the 'ort' strategy.

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git branch -d hotfix
Deleted branch hotfix (was d165cf1).

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git reflog
5633428 (HEAD -> master) HEAD@{0}: merge hotfix: Merge made by the 'ort' strategy.
81088ce (origin/master) HEAD@{1}: checkout: moving from hotfix to master
d165cf1 HEAD@{2}: checkout: moving from master to hotfix
81088ce (origin/master) HEAD@{3}: pull origin master --rebase (finish): returning to refs/heads/master
81088ce (origin/master) HEAD@{4}: pull origin master --rebase (pick): a added in release
7f421d0 HEAD@{5}: pull origin master --rebase (pick): entered
d6e7854 HEAD@{6}: pull origin master --rebase (start): checkout d6e7854452bc137cf7d2973b0ab053f817555da7
691e669 HEAD@{7}: checkout: moving from release to master
1074366 (release) HEAD@{8}: pull origin master --rebase (finish): returning to refs/heads/release
1074366 (release) HEAD@{9}: pull origin master --rebase (pick): a added in release
d6e7854 HEAD@{10}: pull origin master --rebase (start): checkout d6e7854452bc137cf7d2973b0ab053f817555da7
3693282 (origin/release) HEAD@{11}: checkout: moving from master to release
691e669 HEAD@{12}: merge release: Merge made by the 'ort' strategy.
3fcb2f0 HEAD@{13}: checkout: moving from release to master
3693282 (origin/release) HEAD@{14}: checkout: moving from 44c60abda5c91adfcc83c272f55f8ed219c40c0d to release
44c60ab HEAD@{15}: commit: Updated alphabet file
f990a92 HEAD@{16}: rebase (pick): added n=in release
3c4bf46 HEAD@{17}: rebase (start): checkout HEAD~4
3fcb2f0 HEAD@{18}: commit: entered
```

```

8693282 (origin/release) HEAD@{11}: checkout: moving from master to release
591e669 HEAD@{12}: merge release: Merge made by the 'ort' strategy.
8fcb2f0 HEAD@{13}: checkout: moving from release to master
8693282 (origin/release) HEAD@{14}: checkout: moving from 44c60abda5c91adfcc83c272f55f8ed219c40c0d
to release
44c60ab HEAD@{15}: commit: Updated alphabet file
f990a92 HEAD@{16}: rebase (pick): added n=in release
8c4bf46 HEAD@{17}: rebase (start): checkout HEAD~4
8fcb2f0 HEAD@{18}: commit: entered
07e764b HEAD@{19}: commit (amend): Created new message
995d6a7 HEAD@{20}: commit (amend): New Message
1fca2e5 HEAD@{21}: commit (amend): New Message
d6e7854 HEAD@{22}: reset: moving to HEAD
d6e7854 HEAD@{23}: commit: a added in master
1ae2f1d HEAD@{24}: checkout: moving from release to master

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git checkout -b hotfix d165cf1
warning: unable to rmdir 'techie-05': Directory not empty
Switched to a new branch 'hotfix'

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git branch
* hotfix
  master
  release

```

4. Undo Wrong Push

```

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git status
On branch hotfix
Your branch is up to date with 'origin/hotfix'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
  (commit or discard the untracked or modified content in submodules)
        deleted:    file
        modified:   techie-05 (untracked content)

no changes added to commit (use "git add" and/or "git commit -a")
Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git add .

```

```

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git commit -m "deleting file"
[hotfix c62f2c3] deleting file
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 file

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ ls
hello hotfix release techie-05/ test

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git push -u origin hotfix
Enumerating objects: 3, done.

```

```

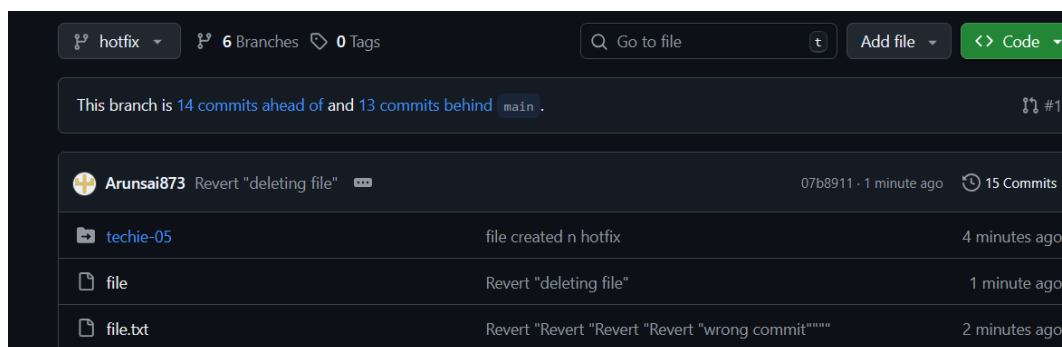
Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git revert c62f2c
[hotfix 07b8911] Revert "deleting file"
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file

Arunsaï@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ ls
file file.txt hello hotfix release techie-05/ test

```

5. Push a wrong commit to GitHub, then undo it without losing history.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git push -u origin hotfix
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 556 bytes | 556.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/Arunsai873/techie-05.git
   c62f2c3..07b8911 hotfix -> hotfix
branch 'hotfix' set up to track 'origin/hotfix'.
```



Amend a Commit

1. Make a commit, then add a missing file to it using git commit --amend.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ ls
file file.txt file1.txt hello hotfix release techie-05/ test
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git add .
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git commit -m "Initial commit"
[hotfix d08617a] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 file1.txt
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git add file2.txt
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git commit --amend
[hotfix f6e09b5] Initial commit
Date: Thu Apr 9 15:16:41 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
```

2. Take a specific commit from one branch and apply it to another branch.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git log -2
commit 47aff28bf60aa802e3a0e283f56911b26c40a929 (HEAD -> hotfix, origin/hotfix)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Thu Apr 9 15:46:29 2026 +0530

    added javeed

commit f6e09b54c7bc073d89e30bcdeedf95ae36c0e4b9
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Thu Apr 9 15:16:41 2026 +0530

    Initial commit
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (hotfix)
$ git checkout master
Switched to branch 'master'
M   techie-05
Your branch is ahead of 'origin/master' by 2 commits.
    (use "git push" to publish your local commits)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git cherry-pick 47aff28
[master db3a387] added javeed
Date: Thu Apr 9 15:46:29 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 javeed
```

3. Reorder and squash multiple commits into a single clean commit.

```
no changes added to commit (use "git add" and/or "git commit -a")

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|REBASE 1/4)
$ git rebase --continue
[detached HEAD e424c29] a added in release
Date: Wed Apr 8 17:43:49 2026 +0530
3 files changed, 1 deletion(-)
create mode 100644 file.txt
create mode 100644 javeed
Successfully rebased and updated refs/heads/master.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .
```

```
pick d165cf1 added in hotfix
squash 1c09fd9 alphabet
squash 8aa4a5a added
squash db3a387 added javeed

# Rebase 81088ce..db3a387 onto 81088ce (4 commands)
#
#
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log -3
commit e424c2997201d18b1869b9b2c78e8dc6b116696b (HEAD -> master, origin/master)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 17:43:49 2026 +0530

    a added in release

    alphabet

    added

    added javeed
```

4. Create a version tag (v1.0), push it to GitHub, then delete and restore it.

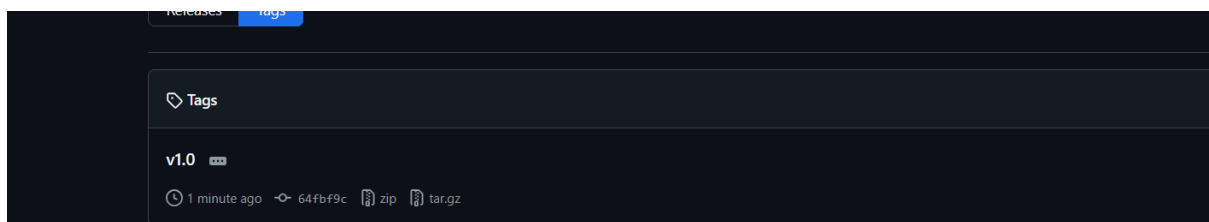
```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git tag v1.0

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git tag
1.0
v1.0

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push origin v1.0
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git tag v1.0 64fbf9c -m "Restored v1.0 tag"

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push origin v1.0
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 405 bytes | 405.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Arunsai873/techie-05.git
* [new tag]          v1.0 -> v1.0
```



5. Clone only a subdirectory of a repo using sparse checkout.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git clone --no-checkout git@github.com:Arunsai873/techie-05.git techie-sparse
Cloning into 'techie-sparse'...
remote: Enumerating objects: 110, done.
remote: Counting objects: 100% (110/110), done.
remote: Compressing objects: 100% (70/70), done.
remote: Total 110 (delta 34), reused 75 (delta 20), pack-reused 0 (from 0)
Receiving objects: 100% (110/110), 18.01 KiB | 658.00 KiB/s, done.
Resolving deltas: 100% (34/34), done.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cd techie-sparse

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-sparse (main)
$ ls

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-sparse (main)
$ git sparse-checkout init --cone

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-sparse (main|SPARSE)
$ git sparse-checkout set devops

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-sparse (main|SPARSE)
$ git checkout master
Switched to a new branch 'master'
branch 'master' set up to track 'origin/master'.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/techie-sparse (master|SPARSE)
$ ls
a .txt alphabet b.txt devops/ file.txt hello hotfix javeed master release techie techie-05/
```


Reset vs Revert Challenge

1. Demonstrate the difference between git reset --hard and git revert in a repo.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log --oneline
64fbf9c (HEAD -> master, tag: v1.0) Revert "a added in master"
e424c29 (tag: 1.0, origin/master) a added in release
7f421d0 entered
d6e7854 a added in master
1ae2f1d two files added
7c2a160 added
3c4bf46 added file techie in techie
da85fde added file techie in release
34ad57b committed in merge
f430e34 Added alphabet file
0a1c500 alphabet
5f133cd hello
83d91b0 changed the commit message using ammend command
c433502 added n=in release
ed1ed72 added file in master
e49c7d2 added hello in master
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git reset --hard HEAD~1
HEAD is now at e424c29 a added in release

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log --oneline
e424c29 (HEAD -> master, tag: 1.0, origin/master) a added in release
7f421d0 entered
d6e7854 a added in master
1ae2f1d two files added
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git revert HEAD
[master 2a595d8] Revert "a added in release"
 3 files changed, 1 insertion(+)
 delete mode 100644 file.txt
 delete mode 100644 javeed

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log --oneline
2a595d8 (HEAD -> master) Revert "a added in release"
e424c29 (tag: 1.0, origin/master) a added in release
7f421d0 entered
d6e7854 a added in master
```

2. Checkout a specific commit (detached HEAD state) and create a new branch from it.


```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log --oneline
2a595d8 (HEAD -> master) Revert "a added in release"
e424c29 (tag: 1.0, origin/master) a added in release
7f421d0 entered
d6e7854 a added in master
1ae2f1d two files added
7c2a160 added
3c4bf46 added file techie in techie
da85fde added file techie in release
34ad57b committed in merge
f430e34 Added alphabet file
0a1c500 alphabet
5f133cd hello
83d91b0 changed the commit message using ammd command
c433502 added n=in release
ed1ed72 added file in master
e49c7d2 added hello in master

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git checkout 3c4bf46
Note: switching to '3c4bf46'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

```

```

HEAD is now at 3c4bf46 added file techie in techie

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app ((3c4bf46...))
$ git checkout -b pavan
Switched to a new branch 'pavan'

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ ls
alphabet hello hotfix master techie techie-sparse/

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$

```

Git Hooks Challenge

1. Configure a pre-commit hook to reject commits without a message format (e.g., must start with JIRA-XXX).

```

GNU nano 6.3 commit-msg
chmod +x commit-msg#!/bin/bash

commit_msg_file=$1
commit_msg=$(cat "$commit_msg_file")

if [[ ! $commit_msg =~ ^JIRA-[0-9]+ ]]; then
    echo "✗ Commit message must start with JIRA-XXX (e.g., JIRA-123 Fix bug)"
    exit 1
fi

```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ nano .git/hooks/commit-msg

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ nano .git/hooks/commit-msg

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ chmod +x .git/hooks/commit-msg

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ git add .

```

```

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ git commit -m "JIRA-101 fixed bug"
[pavan 875c11e] JIRA-101 fixed bug
1 file changed, 1 insertion(+)
create mode 160000 techie-sparse

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ git commit -m "JIRA-123 fixed bug"
On branch pavan
nothing to commit, working tree clean

```

```

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ cat .git/hooks/commit-msg
#!/bin/bash
commit_msg_file=$1
commit_msg=$(cat "$commit_msg_file")

if [[ ! $commit_msg =~ ^JIRA-[0-9]+ ]]; then
    echo "X Commit message must start with JIRA-XXX (e.g., JIRA-123 Fix bug)"
    exit 1
fi

```

2. Show the difference between squash merge and rebase merge with evidence.

```

no changes added to commit (use "git add" and/or "git commit -a")

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master|REBASE 1/4)
$ git rebase --continue
[detached HEAD e424c29] a added in release
Date: Wed Apr 8 17:43:49 2026 +0530
3 files changed, 1 deletion(-)
create mode 100644 file.txt
create mode 100644 javeed
Successfully rebased and updated refs/heads/master.

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .

```

```

pick d165cf1 added in hotfix
squash 1c09fd9 alphabet
squash 8aa4a5a added
squash db3a387 added javeed

# Rebase 81088ce..db3a387 onto 81088ce (4 commands)
#

```

```

Arun sai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log -3
commit e424c2997201d18b1869b9b2c78e8dc6b116696b (HEAD -> master, origin/master)
Author: Arun sai <arunsaimailme17@gmail.com>
Date: Wed Apr 8 17:43:49 2026 +0530

    a added in release

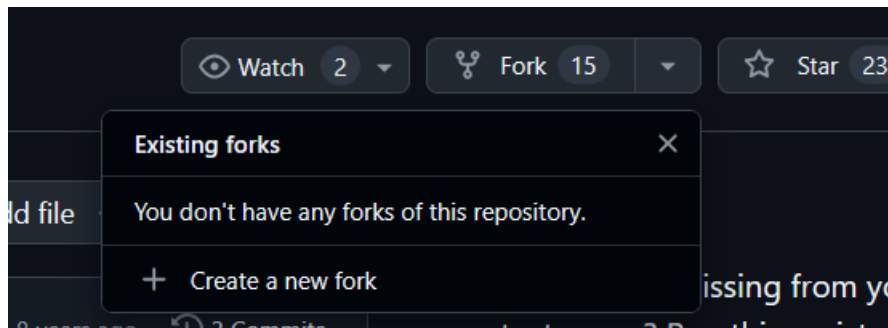
    alphabet

    added

    added javeed

```

3. Fork a repo, make a change, and submit a pull request to the original repo.



Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk ().*

Owner * Arunsai873 / Repository name * Kumarn
✓ Kumarn is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

88 / 350 characters

☒ Copy the `master` branch only
Contribute back to Hattoff/GitBashHere by adding your own branch. [Learn more.](#)

ⓘ You are creating a fork in your personal account.

[Create fork](#)

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ git clone git@github.com:Arunsai873/Kumarn.git
Cloning into 'Kumarn'...
remote: Enumerating objects: 10, done.
remote: Total 10 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (10/10), done.
Resolving deltas: 100% (2/2), done.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (pavan)
$ cd kumarn
```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (master)
$ git remote -v
origin git@github.com:Arunasai873/Kumarn.git (fetch)
origin git@github.com:Arunasai873/Kumarn.git (push)

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (master)
$ git checkout -b kumar
Switched to a new branch 'kumar'

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ echo "new change" >> file.txt

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches i
t

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git commit -m "JIRA-123 added feature"
[kumar 636fc28] JIRA-123 added feature
1 file changed, 1 insertion(+)
create mode 100644 file.txt

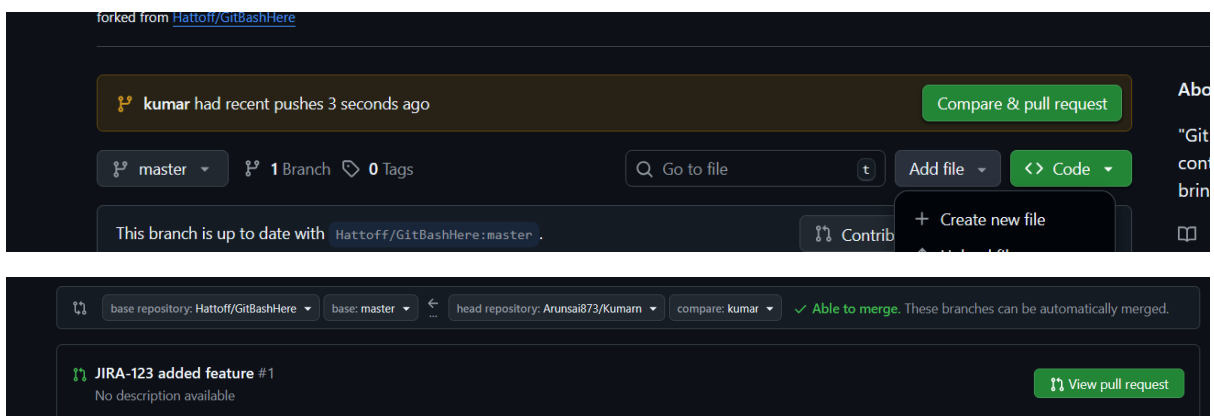
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ echo "new change" >> file.txt

```

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git push origin kumar
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.

```



Recover Lost Commit

1. Commit something, reset hard, and then recover it using git reflog.

```

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ echo "important work" > demo.txt

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git add .
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches i
t

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git commit -m "Important commit"
[kumar 17b812c] Important commit
2 files changed, 2 insertions(+)
create mode 100644 demo.txt

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ echo "important work" > demo.txt

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git log --oneline
17b812c (HEAD -> kumar) Important commit
636fc28 (origin/kumar) JIRA-123 added feature
101b9d7 (origin/master, origin/HEAD, master) Add files via upload
0b5f7a7 Update README.md
ddd12b8 Initial commit

```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git reset --hard HEAD~1
HEAD is now at 636fc28 JIRA-123 added feature
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/kumarn (kumar)
$ git reflog
636fc28 (HEAD -> kumar, origin/kumar) HEAD@{0}: reset: moving to HEAD~1
17b812c HEAD@{1}: commit: Important commit
636fc28 (HEAD -> kumar, origin/kumar) HEAD@{2}: commit: JIRA-123 added fea
101b9d7 (origin/master, origin/HEAD, master) HEAD@{3}: checkout: moving fr
```